PROJECT PROPOSAL

ThreatLens — Cyber-Threat Multitool

A Web-Based Cybersecurity Threat Analysis and Security Assessment Platform

Academic Project Proposal

This proposal presents ThreatLens as a web-based platform for educational, defensive, and security-analysis purposes.

1. Introduction

Cybersecurity has become an essential requirement for organizations, educational institutions, businesses, and individual users due to the increasing number of cyber threats and security incidents. Modern systems face threats such as malicious IP addresses, suspicious URLs, phishing attacks, malware, compromised domains, and other indicators of compromise.

The proposed project, **ThreatLens — Cyber-Threat Multitool**, is a web-based cybersecurity platform designed to provide users with a centralized environment for analyzing potential cyber threats and security indicators.

Instead of relying on multiple separate cybersecurity utilities, ThreatLens aims to bring commonly required threat-analysis capabilities together in a single, user-friendly web interface. The platform can help users understand suspicious digital artifacts and make informed security decisions.

The system is intended primarily for **educational, defensive, and security-analysis purposes**.

2. Problem Statement

Cybersecurity analysts and students often need to use multiple tools and platforms to investigate suspicious indicators such as URLs, IP addresses, domains, files, and other threat-related information.

This creates several problems:

- Security information is distributed across different tools.
- Beginners may find professional cybersecurity tools difficult to understand.
- Manual investigation can be time-consuming.
- It can be difficult to organize and interpret threat information.
- Students have limited opportunities to practice threat-analysis concepts in an interactive environment.

1. Therefore, there is a need for an integrated web-based platform that provides a simple and organized interface for cybersecurity analysis and education.

3. Proposed Solution

ThreatLens proposes a centralized **Cyber-Threat Multitool** that allows users to submit security-related indicators and analyze them through an interactive web interface.

The platform can organize analysis into different modules, allowing users to investigate suspicious information and understand potential security risks.

The system focuses on:

- Threat identification
- Indicator analysis
- Security assessment
- Threat intelligence concepts
- Suspicious URL/domain analysis
- IP-related investigation
- Security awareness
- Educational cybersecurity analysis

1. The interface is designed to make cybersecurity concepts easier to understand while maintaining a professional security-dashboard experience.

4. Objectives

The major objectives of ThreatLens are:

- Develop a centralized web-based cybersecurity analysis platform.
- Provide an easy-to-use interface for threat investigation.
- Allow users to analyze security indicators from a single platform.
- Present cybersecurity results in an understandable format.
- Help students understand practical threat-intelligence concepts.
- Improve awareness of common cyber threats.
- Provide a foundation for defensive cybersecurity analysis.

1. Reduce the need to switch between multiple analysis interfaces.
2. Organize security information systematically.
3. Provide visual representations of security findings.
4. Improve the learning experience for cybersecurity students.
5. Create a scalable architecture for adding future security modules.

5. Scope of the Project

The scope of ThreatLens includes the development of a web-based cybersecurity analysis environment.

In Scope

- URL analysis
- Domain analysis
- IP address investigation
- Threat indicators
- Suspicious activity analysis
- Security scoring
- Threat classification
- Security reports
- Dashboard visualization
- Educational explanations
- Analysis history

1. The system is intended to support **authorized and defensive security analysis**.

Out of Scope

- Unauthorized access to systems
- Credential theft
- Malware deployment
- Attacking third-party systems
- Bypassing security controls

- Unauthorized exploitation

Its purpose is security analysis, education, awareness, and defensive investigation.

6. Target Users

ThreatLens can be useful for several categories of users.

6.1 Cybersecurity Students

Students can use the platform to learn how security indicators are investigated in real-world cybersecurity environments.

6.2 Security Researchers

Researchers can use the platform as an interface for organizing threat-analysis activities.

6.3 SOC Analysts

Security Operations Center analysts can potentially use similar functionality for preliminary investigation and indicator triage.

6.4 IT Professionals

IT administrators can use threat-analysis concepts to investigate potentially suspicious infrastructure.

6.5 Cybersecurity Educators

Teachers and instructors can use the platform for demonstrations and practical cybersecurity exercises.

7. Major System Features

7.1 Cybersecurity Dashboard

- Total analyses
- Threat indicators
- Risk levels
- Recent investigations
- Security statistics
- Analysis categories

7.2 URL Analysis

- URL structure
- Domain information
- Suspicious characteristics
- Risk indicators
- Security classification

7.3 IP Address Analysis

- IP address
- Geographic information
- Network information

- Reputation information
- Risk classification
- Associated indicators

7.4 Domain Analysis

- Domain information
- Domain reputation
- Registration-related information
- Security indicators
- Risk assessment

7.5 Threat Intelligence

- Malicious
- Suspicious
- Low Risk
- Medium Risk
- High Risk
- Unknown

7.6 Risk Scoring

Low Risk → Medium Risk → High Risk → Critical

7.7 Security Reports

- Submitted indicator
- Analysis type
- Findings
- Risk level
- Security recommendations
- Timestamp
- Overall assessment

7.8 Analysis History

- Previous indicators
- Analysis dates
- Results
- Risk classifications
- Investigation types

8. System Workflow

User → ThreatLens Dashboard → Select Analysis Module → Submit Security Indicator → Input Validation → Threat Analysis → Threat Intelligence / Security Data → Risk Assessment → Result Visualization → Security Recommendation → Generate / Review Report

9. System Architecture

The proposed architecture can consist of three primary layers.

9.1 Presentation Layer

The frontend provides the user interface.

- HTML
- CSS
- JavaScript
- React / Next.js
- Responsive UI components

9.2 Application Layer

The application layer handles core application processing.

- User requests
- Input validation
- Analysis logic
- Risk scoring
- API communication
- Report generation

9.3 Data / Intelligence Layer

The architecture can be expanded as additional cybersecurity services are integrated.

- Threat intelligence sources
- Security APIs
- Analysis databases
- Indicator information
- Historical analysis data

10. Proposed Technologies

Frontend

- HTML5
- CSS3
- JavaScript
- React.js / Next.js

Backend

- Node.js
- Python
- REST APIs

Database

- PostgreSQL
- MySQL
- MongoDB

Deployment

- The current web project is deployed through Vercel, providing a web-accessible interface for the application.

11. Security Considerations

Since ThreatLens is itself a cybersecurity application, security must be considered throughout development.

Important security controls include:

- Input validation
- Secure API communication
- HTTPS
- Authentication and authorization
- Rate limiting
- Secure API-key storage
- Protection against injection attacks
- Secure session management
- Error handling
- Logging and monitoring

1. Sensitive credentials and API keys should never be exposed in frontend source code.

12. User Interface Design

The interface should follow a modern cybersecurity dashboard design.

Important UI elements can include:

- Dark security-themed interface
- Dashboard cards
- Threat-level indicators
- Analysis panels
- Search/input fields
- Results tables
- Risk visualization
- Security charts

- Reports
- Navigation sidebar

1. The design should prioritize usability so that both beginners and experienced users can understand the results.

13. Functional Requirements

1. Accept security indicators from users.
2. Validate submitted input.
3. Process analysis requests.
4. Communicate with relevant security intelligence services.
5. Analyze returned information.
6. Calculate or determine risk levels.
7. Display results.
8. Store analysis history where applicable.
9. Generate security reports.
10. Provide recommendations based on findings.

14. Non-Functional Requirements

Performance

The application should respond efficiently to user requests.

Usability

The interface should be simple and intuitive.

Scalability

The architecture should allow additional cybersecurity modules to be added.

Reliability

The application should handle invalid inputs and API failures gracefully.

Security

User data and API credentials should be protected.

Maintainability

The code should be modular and documented.

15. Advantages of ThreatLens

- Centralized cybersecurity analysis
- User-friendly interface
- Educational value
- Faster preliminary investigation
- Security risk visualization

- Modular architecture
- Expandable functionality
- Practical cybersecurity learning environment

16. Innovation

The primary innovation of ThreatLens is the integration of multiple cybersecurity-analysis concepts into a single web-based environment.

Instead of presenting cybersecurity tools as isolated utilities, the platform attempts to create a unified workflow:

Indicator → **Analysis** → **Intelligence** → **Risk Assessment** → **Visualization** → **Report**

This workflow can provide students with a practical understanding of how security analysts approach suspicious indicators.

17. Future Enhancements

- User authentication
- Role-based access control
- Advanced threat-intelligence integration
- Malware hash analysis
- File reputation analysis
- Phishing detection
- Email security analysis
- Threat correlation
- MITRE ATT&CK; mapping
- IOC relationship visualization
- Automated security reports
- SIEM integration
- SOC dashboard
- Alert management
- Machine-learning-based risk classification

A future version could potentially evolve into a lightweight **Security Operations Center (SOC) investigation platform**.

18. Expected Outcomes

At the completion of the project, ThreatLens is expected to provide a functional web-based cybersecurity analysis platform capable of demonstrating practical threat-investigation concepts.

Expected outcomes include:

- Functional cybersecurity dashboard
- Multiple analysis modules
- Threat-risk visualization
- Security recommendations

- Analysis history
- Structured reporting
- Educational cybersecurity functionality
- Scalable application architecture

19. Testing Strategy

Functional Testing

Verify that every module performs its intended function.

UI Testing

Ensure that the interface works correctly across different screen sizes.

Security Testing

Test input validation, authentication, API security, and common web vulnerabilities.

Performance Testing

Evaluate response times and application behavior under different loads.

API Testing

Verify successful and unsuccessful responses from integrated threat-intelligence services.

User Acceptance Testing

Allow students or intended users to test the platform and provide feedback.

20. Project Deliverables

1. Functional ThreatLens web application
2. Source code
3. Database/API configuration
4. Project documentation
5. System architecture diagram
6. Flowcharts
7. Testing documentation
8. User manual
9. Final project report
10. Presentation/demo

21. Conclusion

ThreatLens — Cyber-Threat Multitool is proposed as a web-based cybersecurity analysis platform that brings multiple threat-investigation concepts into a centralized and user-friendly environment.

The project addresses the difficulty of understanding and analyzing cybersecurity indicators by providing a structured workflow for investigation, risk assessment, visualization, and reporting.

The platform has strong educational value because it allows cybersecurity students to understand practical concepts used in defensive security and threat intelligence.

With further development and integration of additional intelligence sources, ThreatLens could be expanded from an academic cybersecurity project into a more comprehensive defensive security-analysis platform.

22. Project Summary

Category	Description
Project Name	ThreatLens
Project Type	Web-Based Cybersecurity Platform
Domain	Cybersecurity / Threat Intelligence
Primary Purpose	Threat Analysis & Security Assessment
Target Users	Students, Researchers, IT & Security Professionals
Platform	Web Application
Frontend	React / Next.js / Web Technologies
Backend	Node.js / Python / APIs
Database	PostgreSQL / MongoDB / MySQL
Deployment	Web / Vercel
Security Focus	Defensive & Educational
Future Scope	SOC, SIEM, Threat Intelligence & ML

Proposed Project Tagline

“See the Threat. Understand the Risk. Defend Smarter.”